

Prompt Engineering — Quiz 5

Name: _____ Date: _____

Instructions: This quiz contains 12 multiple-choice questions based on the "Prompting for Complex Problem Solving" lecture. Answer all questions.

#1 What is the primary goal of **Decomposition** in the context of solving complex problems with LLMs?

- A) To reduce the number of tokens used in the prompt.
- B) To break down ambiguous or large problems into manageable, promptable components.
- C) To force the model to use external tools like a calculator.
- D) To increase the temperature setting of the model.

#2 Which prompting technique encourages the model to "show its work" or reasoning steps before providing a final answer?

- A) Zero-Shot Prompting
- B) Chain-of-Thought (CoT)
- C) Few-Shot Prompting
- D) Retrieval Augmented Generation (RAG)

#3 How does the **Tree of Thoughts (ToT)** framework differ from standard linear reasoning?

- A) It forces the model to answer immediately without thinking.
- B) It explores multiple possible reasoning paths or branches simultaneously.
- C) It relies solely on external search engines for answers.
- D) It is only used for creative writing tasks.

#4 In the **ReAct** framework, what are the three core components of the loop?

- A) Input, Process, Output
- B) Thought, Action, Observation
- C) Plan, Execute, Review
- D) Context, Question, Answer

#5 What is the main advantage of **Retrieval Augmented Generation (RAG)**?

- A) It allows the model to run faster on local hardware.
- B) It reduces hallucinations by grounding the model's answer in retrieved external context.
- C) It increases the model's training data permanently.
- D) It removes the need for prompt engineering.

#6 What does the acronym **MECE** stand for in problem structuring?

- A) Most Efficient, Cost Effective
- B) Mutually Exclusive, Collectively Exhaustive

- C) Maximum Effort, Critical Evaluation
- D) Multi-Environment, Cloud Enabled

#7 In the **SCQA** framework, what does the "C" stand for?

- A) Context
- B) Complication
- C) Conclusion
- D) Cost

#8 What is the first step in **Least-to-Most Prompting**?

- A) Solve the hardest part of the problem first.
- B) Decompose the problem into a list of sub-questions.
- C) Ask the model to guess the final answer.
- D) Provide a dictionary of terms.

#9 How does **Self-Consistency** improve the reliability of LLM outputs?

- A) By asking the model to answer the same question multiple times and picking the most frequent answer.
- B) By restricting the model to a single possible answer.
- C) By increasing the model's vocabulary size.
- D) By using a larger context window.

#10 What is the purpose of **Adversarial Prompting** (or Red Teaming) your own plan?

- A) To make the model angry.
- B) To find logical gaps, risky assumptions, and flaws in the proposed solution.
- C) To generate more polite responses.
- D) To translate the plan into another language.

#11 Why might an LLM need to use an external tool like a **Calculator**?

- A) LLMs are probabilistic text predictors and often struggle with precise arithmetic.
- B) LLMs cannot process numbers at all.
- C) Calculators are faster than LLMs at generating text.
- D) To save battery power on the server.

#12 What is a common issue caused by **Greedy Decoding** in LLMs when planning?

- A) The model refuses to answer.
- B) The model picks the most likely *next* word without considering the long-term goal.
- C) The model generates text in the wrong language.
- D) The model crashes due to memory errors.